Support Vector Machine Analysis - Student Dropout Prediction

Understanding the Pathways to Academic Achievement

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Contents

1 Support Vector Machine Analysis

We will now test different vector models to understand what is the best classifier for our student's performance. SVMs operate by identifying where the **margin** between different categories of data is the greatest. To classify our dataset, the model plots a separating line between the two outcome classes using a number of predictors - in order to keep it clear, we want to create a two-dimensional plane (two predictors).

By applying different kernels—linear, radial, and polynomial— we assess how the shape of the decision boundary affects prediction accuracy and determine which kernel provides the most reliable classification for our dataset.

1.1 Feature Selection

For us to choose the best predictors, we will compute **F-statistic ANOVA** for every numeric feature in the dataset. The F-score tell us *how much* the mean of each feature differs between the **two target groups** (*Graduation/Dropout*). higher scores mean there is a bigger separation between classes, and a better cutoff between groups. Results indicate that grades and the amount of credits approved have the highest F-Values, suggesting they are the most significant predictors for predicting the success of graduating or not.

In this case, we want to choose the features of **second semester marks** and the **amount of units passed** to plot our model.

Table 1: Top 5 Features by F-statistic

	Feature	F_value	p_value
31	Curricular.units.2nd.sem..approved.	2711.44	0
32	Curricular.units.2nd.sem..grade.	2098.45	0
25	Curricular.units.1st.sem..approved.	1613.96	0
26	Curricular.units.1st.sem..grade.	1344.07	0
17	Tuition.fees.up.to.date	881.55	0

1.2 Prepare Data for SVM

Support Vector Machines require **predictors** (numerical variables) and a **target** (categorical: *Graduation/Dropout*) for classification. The data was then structured into a format suitable for SVM modeling. The original dataset contained three academic outcomes: *Enrolled*, *Dropout*, and *Graduate*. Since the objective of this analysis is to distinguish students who ultimately graduate from those who drop out, **the Enrolled category was removed**. This reduces the dataset size but produces a cleaner binary classification.

Table 2: Dataset Summary

Metric	Value
Total Observations	3630
Target Classes	Dropout, Graduate

1.3 Exploratory Visualization

1.4 Train/Test Split

Now, we will split the data into training and testing sets, and then fit several SVM models to classify the students based on their marks.

Table 3: Train/Test Split

Dataset	Observations	Percentage
Training	2905	80%
Testing	725	20%

1.5 Model Training

In our dataset, each student is described by academic features such as second-semester grades and credits passed. If the two groups—graduated and dropout— separate cleanly in a straight line, a linear SVM finds the optimal boundary that splits high-performing students from those with a higher risk of dropping out. But if the patterns are more complex, a non-linear kernel like the radial SVM transforms the data so the model can still draw a clear dividing boundary. This allows the SVM to classify students accurately even when the relationship between performance and graduation is not linear.

Linear Kernel: Creates a straight decision boundary to separate classes. This kernel is ideal when the data is linearly separable. In our student dropout analysis, a linear kernel would suggest that the relationship between academic performance metrics and graduation outcomes follows a straightforward, proportional pattern.

Radial Basis Function (RBF) Kernel: Generates curved, non-linear decision boundaries by measuring the distance between data points. This flexible kernel works best at tracking irregular patterns in the data. For our dataset, the RBF kernel can identify students who may succeed despite lower grades in one metric if they compensate with higher performance in another.

Polynomial Kernel: Produces curved boundaries with a specific degree of curve. In student performance prediction, a polynomial kernel might capture scenarios where moderate performance in both metrics leads to different outcomes than extreme performance in either direction, revealing threshold effects or interaction patterns between academic indicators.

1.6 Model Evaluation

Predict and compare the models

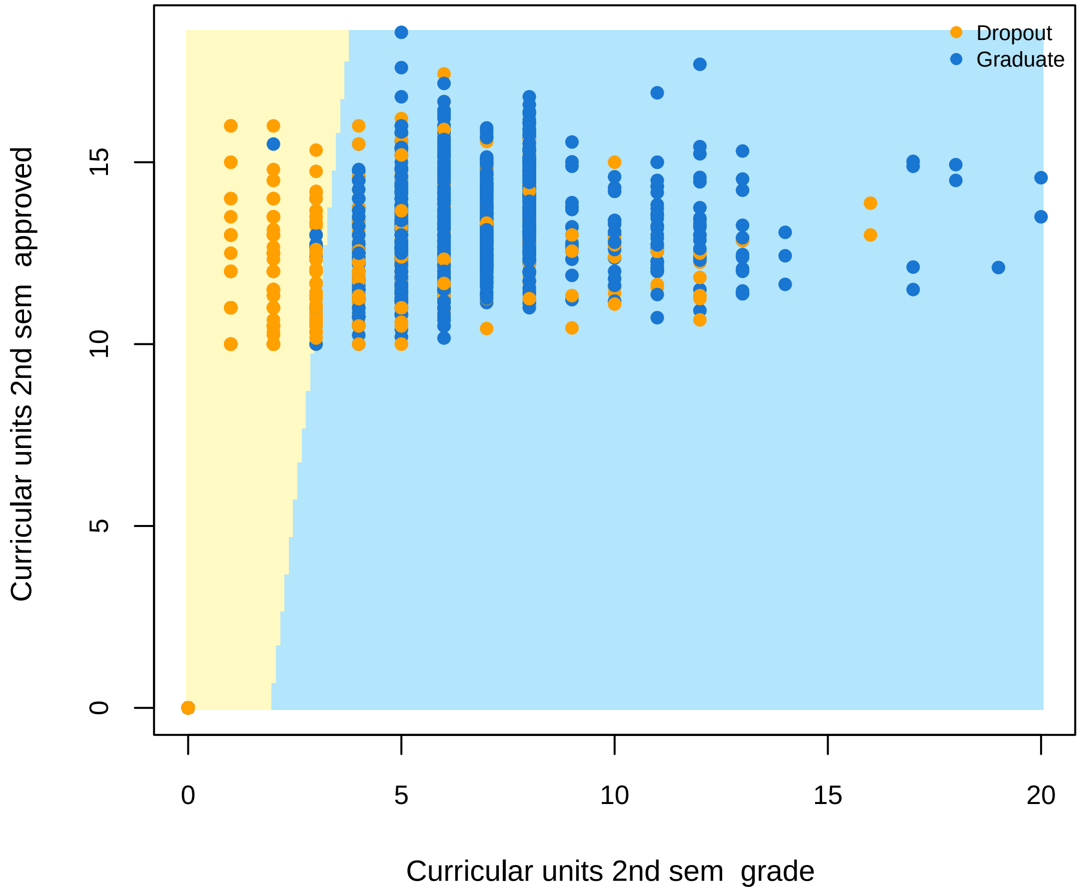
1.6.1 Linear Kernel Results

Table 4: Linear Kernel - Confusion Matrix

	Dropout	Graduate
Dropout	218	25
Graduate	66	416

Table 5: Linear Kernel - Performance Metrics

	Metric	Value
Accuracy	Accuracy	0.8745
Sensitivity	Sensitivity	0.7676
Specificity	Specificity	0.9433
Kappa	Kappa	0.7297



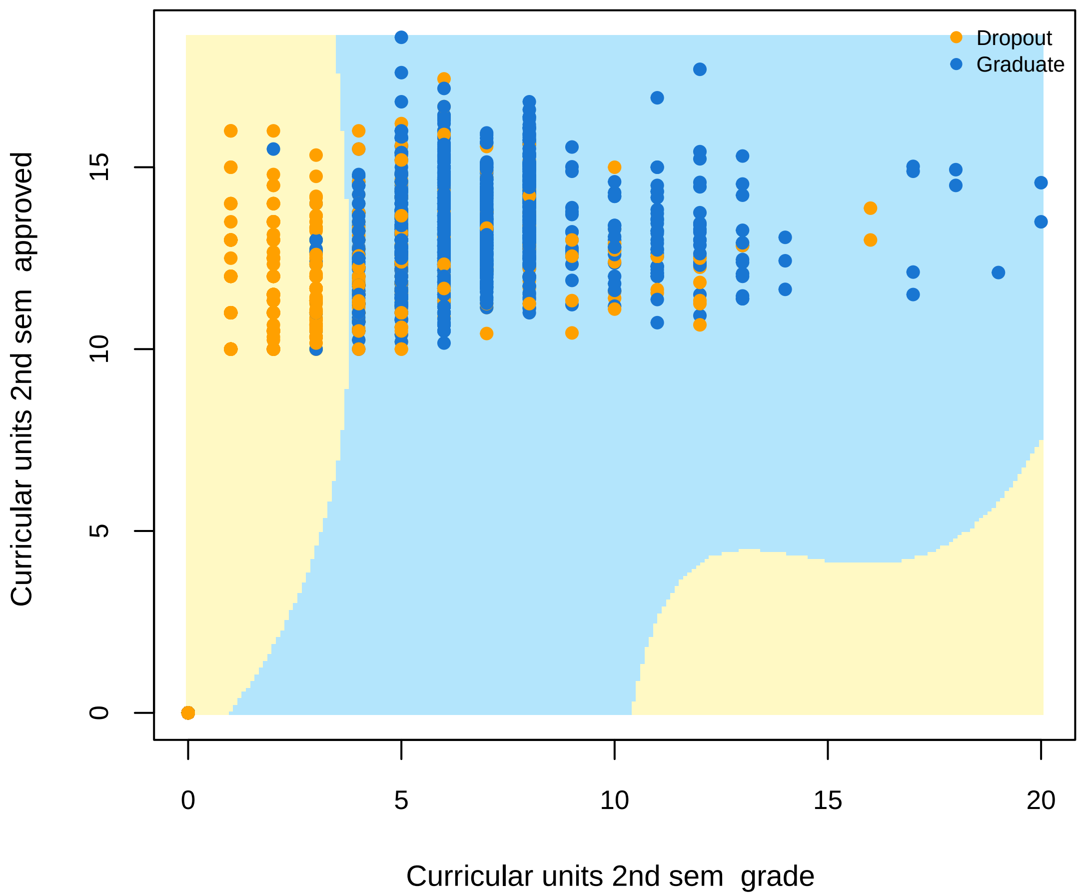
1.6.2 Radial Kernel Results

Table 6: Radial Kernel - Confusion Matrix

	Dropout	Graduate
Dropout	220	25
Graduate	64	416

Table 7: Radial Kernel - Performance Metrics

	Metric	Value
Accuracy	Accuracy	0.8772
Sensitivity	Sensitivity	0.7746
Specificity	Specificity	0.9433
Kappa	Kappa	0.7359



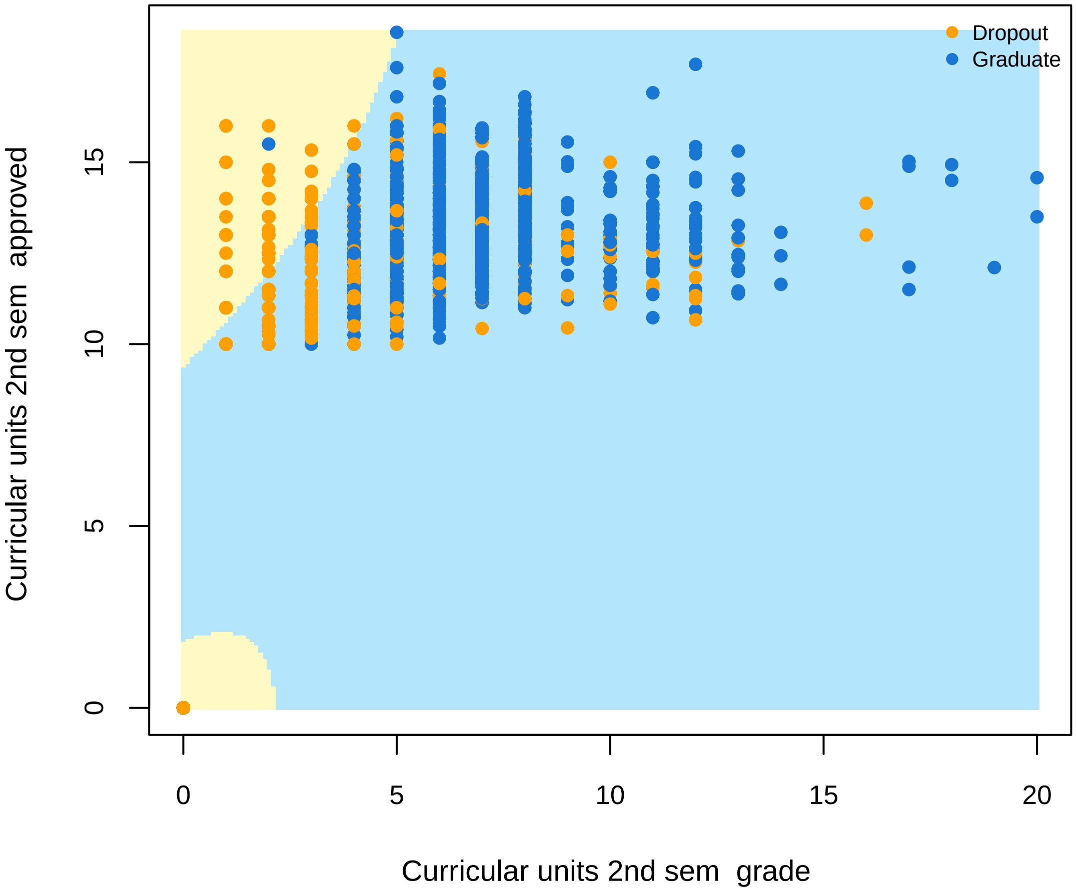
1.6.3 Polynomial Kernel Results

Table 8: Polynomial Kernel - Confusion Matrix

	Dropout	Graduate
Dropout	186	17
Graduate	98	424

Table 9: Polynomial Kernel - Performance Metrics

	Metric	Value
Accuracy	Accuracy	0.8414
Sensitivity	Sensitivity	0.6549
Specificity	Specificity	0.9615
Kappa	Kappa	0.6493



1.7 Performance Comparison

Table 10: Model Accuracy Comparison

Model	Accuracy
Linear	0.874
Radial	0.877
Polynomial	0.841